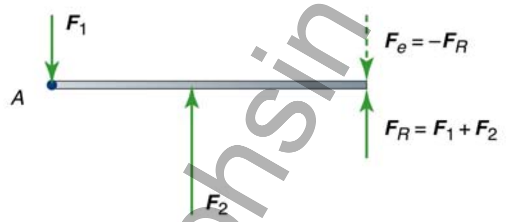
Equilibrium



Equilibrium

EQUATIONS

$$\sum \vec{F} = m\vec{a}$$

$$\sum \vec{M} = I\,\vec{\alpha}$$

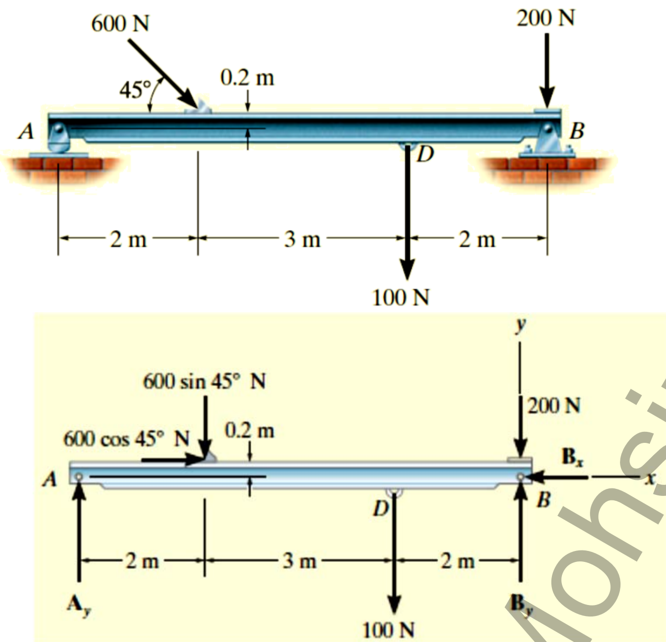
In Equilibrium $a = 0$ and $\alpha = 0$

$$\sum \vec{F} = 0$$

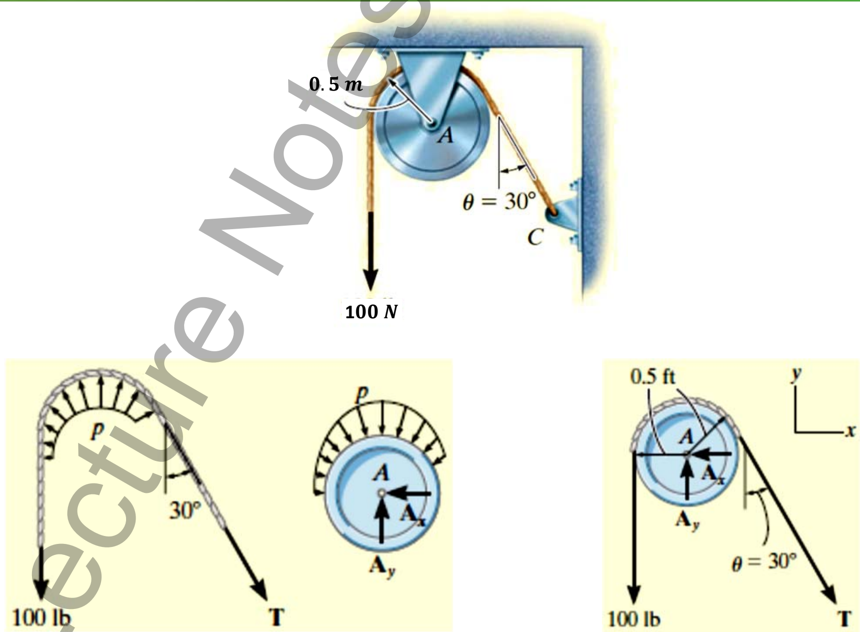
$$\sum \vec{M} = 0$$

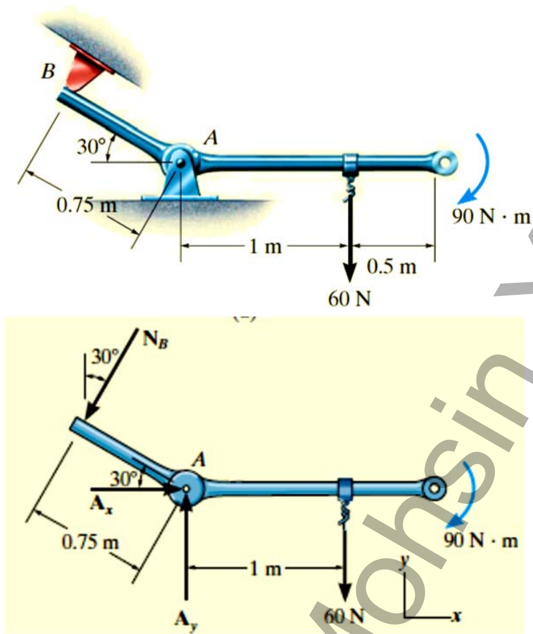
How many equations are there?

six



$$\Sigma F_x = 0$$
$$\Sigma F_y = 0$$
$$\Sigma M_O = 0$$





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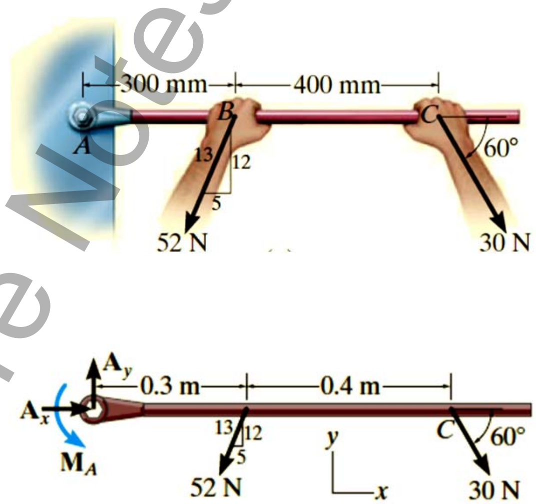

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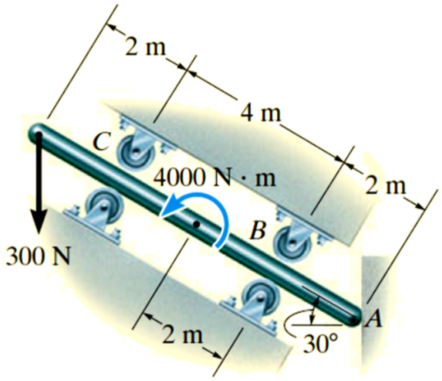
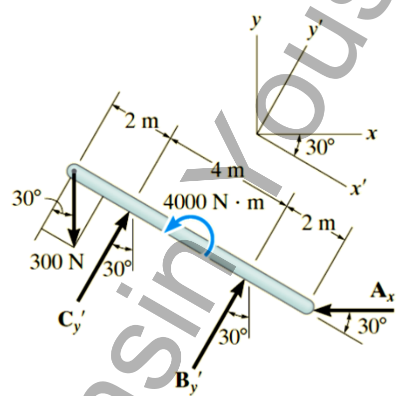
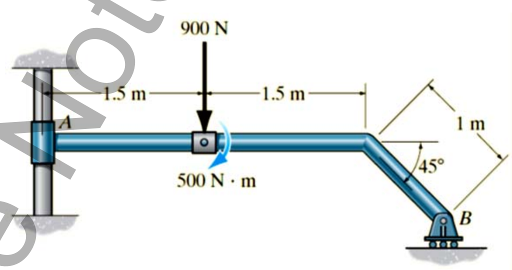
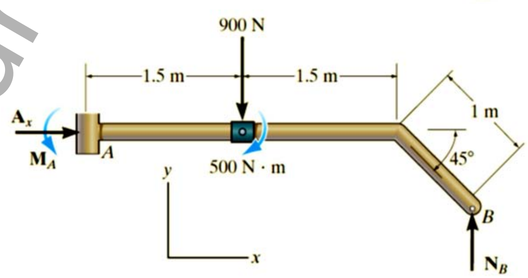
Q. The rod is supported by a smooth wall and at B and C either at the top or bottom by rollers, determine the reactions at these supports. Neglect the weight of the rod.





Q. The collar at A is fixed to the member and can slide vertically along the vertical shaft, determine the support reactions on the member.





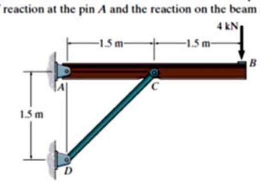
Basic Mech. Engg.

Equilibrium

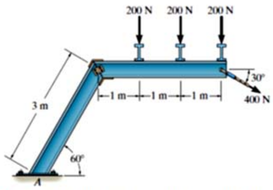
PRACTICE PROBLEMS

Problems 5.2 to 5.6

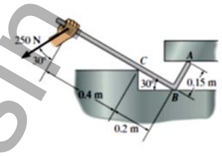
F5–2. Determine the horizontal and vertical components of reaction at the pin *A* and the reaction on the beam at *C*.



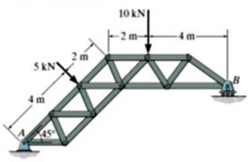
F5–4. Determine the components of reaction at the fixed support *A*. Neglect the thickness of the beam.



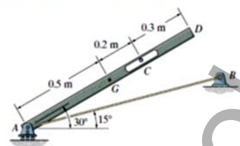
F5–6. Determine the reactions at the smooth contact points *A*, *B*, and *C* on the bar.



F5–3. The truss is supported by a pin at *A* and a roller at *B*. Determine the support reactions.



F5–5. The 25-kg bar has a center of mass at *G*. If it is supported by a smooth peg at *C*, a roller at *A*, and cord *AB*, determine the reactions at these supports.



Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp 226

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Equilibrium

MORE PRACTICE PROBLEMS